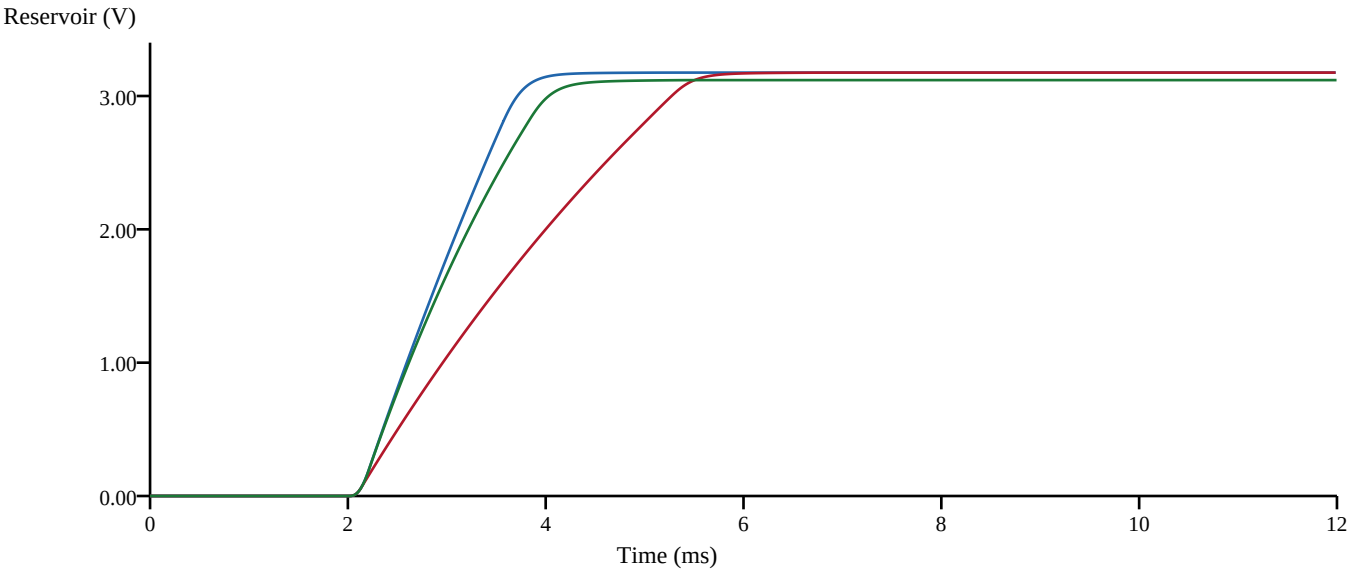
3.3 V source: upper-capacitance startup



startup_3v3_115k_upperC

startup_3v3_210k_upperC

startup_3v3_115k_100mA

All three nominal cases reach a steady reservoir above 3.0 V.

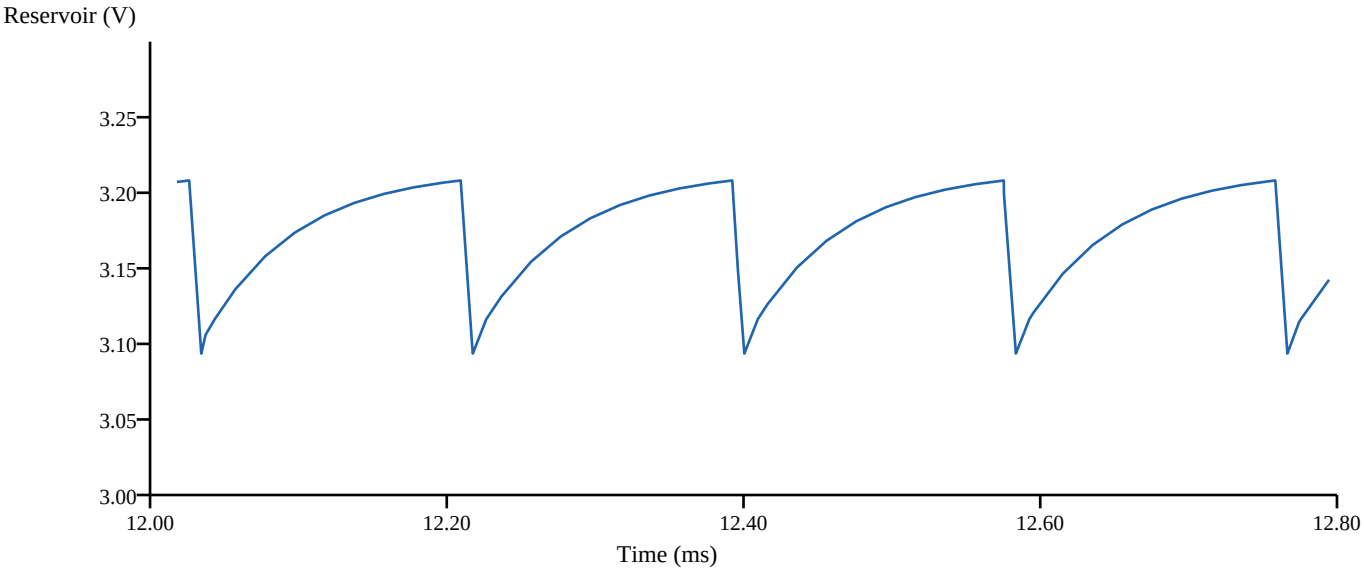
The 210k setting works in this nominal case; it has less current/timing reserve.

Neither trace establishes the complete UCC soft-start or PVT envelope.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.

Imposed bursts: 3.3 V source and conditional 54.38 uF bank



bursts_3v3_115k_lowC

Minimum in the settled full raw trace: 3.0936 V.

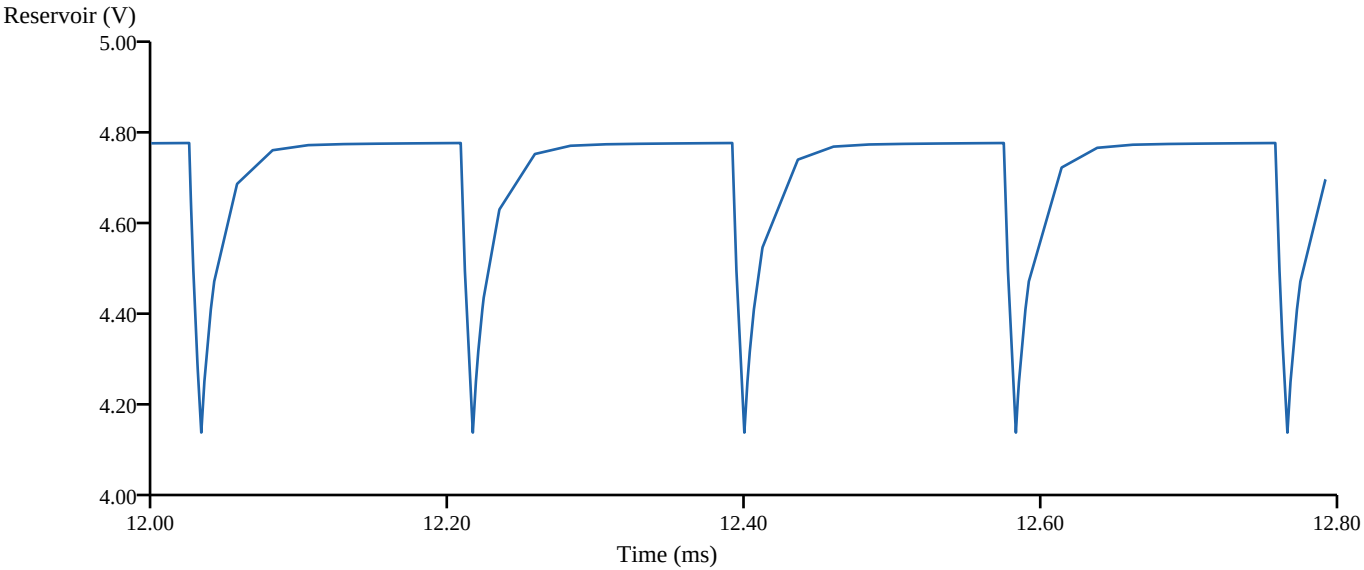
Load is an imposed 0.8 A pulse pattern, not validated UCC behavior.

The stiff-source assumption leaves actual essential-source response open.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.

5 V alternative: imposed bursts with a smaller bank



bursts_5v_115k_smallC

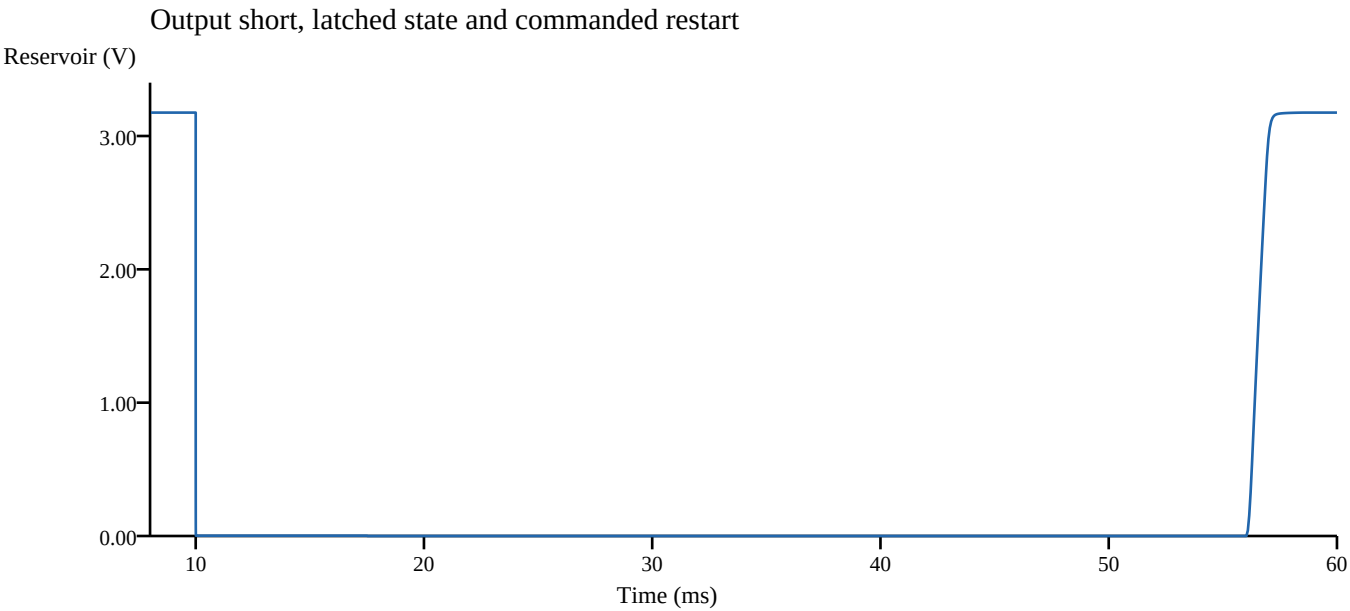
Minimum in the settled full raw trace: 4.1376 V.

The smaller bank permits model current overshoot; RILIM is not a peak bound.

This alternative would make BMS telemetry depend on recovery +5V.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.



short_latch_reset_3v3

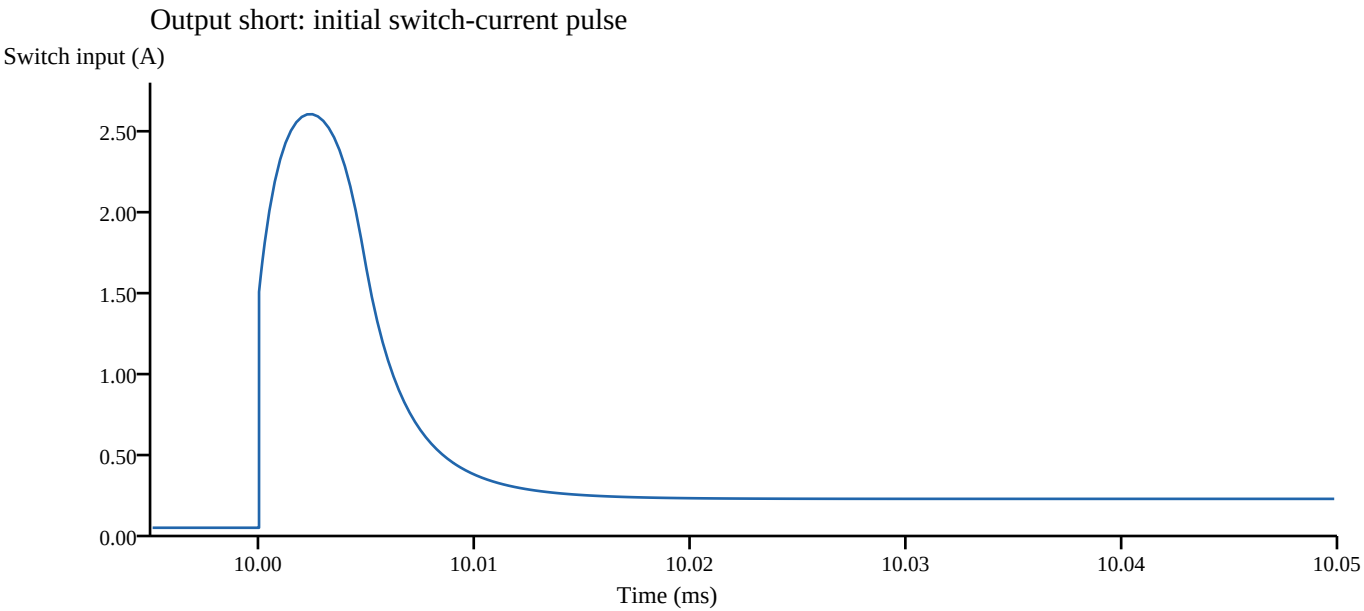
Short applied at 10 ms and removed at 30 ms.

EN low from 35 to 55 ms clears the latch; the output restarts.

This demonstrates latch clearing, not a fully discharged UCC power cycle.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.



short_latch_reset_3v3

Full raw current peak: 2.605 A before limiting.

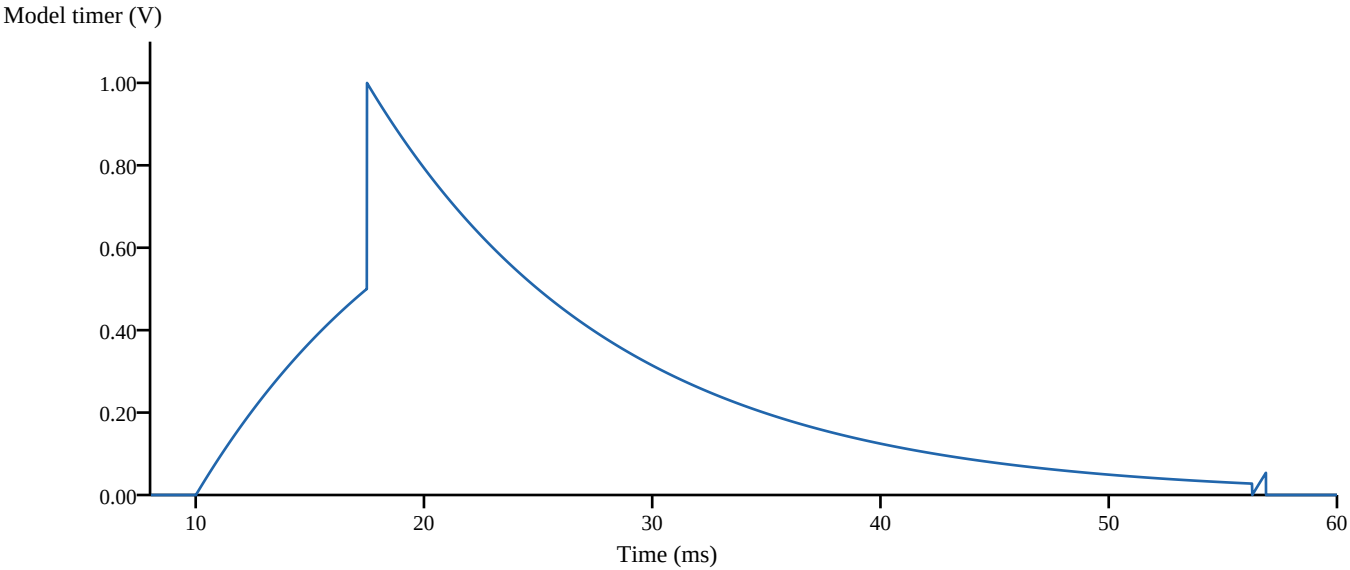
Shared reservoir is imposed at 96 uF behind a stiff 50 milliohm source.

No maximum response-time or MOSFET/resistor survival acceptance follows.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.

Output short: internal model timer and latch



short_latch_reset_3v3

Internal model-node voltage is diagnostic; it is not an external test point.

FAULT is low in the latched interval; EN toggling restores operation.

Temperature, source collapse and a self-consistent UCC load remain open.

TI nominal switch model with XNOR syntax adapter; synthetic source and converter loads.

Decimated plots. Extracted extrema use full raw data. No UCC or system qualification.